CHAPTER 1

Numerical sequences and discrete models

Chapter path

Objectives. Move from a word problem to a sequence; distinguish increment and rate; calculate terms and sums; justify a variation; experiment with a threshold.
Prerequisites. Algebraic manipulation, integer powers, percentages, affine functions, Python loops.
Yearly reference point. 8-17 September: 8 hours, then review

Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Calculate $3 + 2 \times 4$.

2. Calculate 2^0, 2^3 and 2^{-2}.

3. A count of 80 increases by 5%. Give the new count.

4. For $f(x) = 3x - 2$, calculate $f(0)$ and $f(4)$.

5. How many integers are there from 3 to 10 inclusive?

6. A loop `for k in range(4)` takes which values of k ?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. In 90 seconds, explain the difference between index, term and number of terms. Give a counterexample to a rule guessed from three values.

Investigate before formalising

In pairs, compare two collection schemes: 20 kg in the first week, then 5 kg more each week; 20 kg then a 20% increase each week. Conjecture which collects the most in total in six weeks, then justify. Allow 10 min of research and 5 min of comparison.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

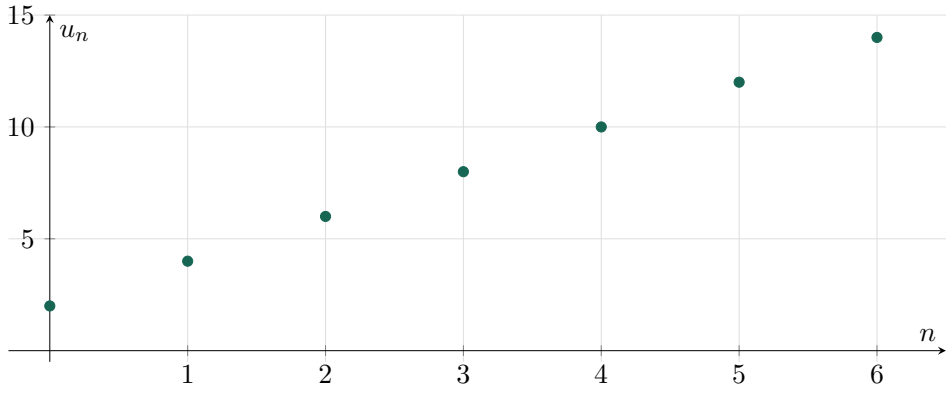
1.1 Defining and representing a sequence

Definition and results

A real sequence is a function whose variable is an integer: to each index n in the domain there corresponds a real number u_n. The notation (u_n) denotes the sequence, u_n a term. An explicit formula gives u_n directly as a function of n. A recursive definition gives a first term and a rule for getting from one term to the next. The representation is a set of points $(n; u_n)$: we do not necessarily join these points.

Status and justification. Definition. The uniqueness of the terms obtained from a first term and of a deterministic rule is assumed; no formal proof by induction is required.

Worked example. For $u_n = n^2 + 1$, $u_0 = 1$, $u_3 = 10$. For $v_0 = 1$ and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. We cannot replace v_3 by $2 \times 3 + 1$: the rule refers to the previous term.



Terms of $u_n = 2n + 2$: only integer indices are plotted.

Watch out

The index is not the value. A finite list of terms does not determine a unique rule.

Immediate application. Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; propose an explicit formula.

1.2 Arithmetic Sequences

Definition and results

A sequence is arithmetic with common difference r if $u_{n+1} = u_n + r$ for every allowed index. Then $u_n = u_p + (n-p)r$ for $n \geq p$. If $r > 0$, it is strictly increasing; if $r < 0$, strictly decreasing; if $r = 0$, constant. The points lie on a line.

Status and justification. For p fixed, the sequence $v_n = u_p + (n-p)r$ satisfies $v_p = u_p$ and $v_{n+1} = v_n + r$. By uniqueness of the successive terms, $u_n = v_n$. The variation follows from the sign of $u_{n+1} - u_n = r$.

Worked example. A reservoir contains 140 L and receives 12 L per minute. After n minutes, $u_n = 140 + 12n$. At 5 min: 200 L. If we start indexing at 1 with 140 L, we write $u_n = 140 + 12(n-1)$.

Watch out

Adding a constant percentage does not in general give an arithmetic sequence.

Immediate application. We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine r and u_0.

1.3 Geometric sequences

Definition and results

A sequence is geometric with common ratio q if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$, with the convention that an empty product equals 1. A constant rate t gives $q = 1 + t$. The variation is determined by $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

Status and justification. The formula gives the first term, and multiplying by q gives the next term. The uniqueness of the generation justifies the formula. For $q = 0$, all terms after the first are zero; deal with this case directly.

Worked example. A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. The terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We get $u_2 = 90$ mg, without confusing a 25% decrease with a 25 mg withdrawal.

Watch out

$q > 1$ is not enough to conclude that every geometric sequence is increasing: the sign of the first term matters.

Immediate application. Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

1.4 Sums, indices and proofs

Definition and results

For $n \geq 1$, $1+2+\cdots+n = \dfrac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1+q+\cdots+q^n = \dfrac{1-q^{n+1}}{1-q}$. For $q = 1$, the sum is $n+1$. Thus, from u_p to u_n, an arithmetic sum equals $\dfrac{(n-p+1)(u_p+u_n)}{2}$; a geometric sum is $\dfrac{u_p(1-q^{n-p+1})}{1-q}$ if $q \neq 1$.

Status and justification. Writing $S = 1 + \cdots + n$ in both directions, each pair of the sum 2 S is $n+1$, with n pairs. For $T = 1 + q + \cdots + q^n$, subtract $qT = q + \cdots + q^{n+1}$: $(1-q)T = 1 - q^{n+1}$. The case $q = 1$ must come before the division.

Worked example. For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \dfrac{5(7+15)}{2}$ = 55. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \dfrac{2(1-3^4)}{1-3} = 80$.

Watch out

Count the terms before using the formula: from 0 to n, there are n+1.

Immediate application. Calculate $\sum_{k=2}^{5} 2^k$ without a summation symbol in the answer.

1.5 Conjecture a limit and program a threshold

Definition and results

A limit is approached intuitively here. The terms $\dfrac{1}{n+1}$ approach 0; n^2 grows without bound; $(-1)^n$ does not settle down. A numerical experiment suggests a conclusion but does not prove a behaviour for all indices. A threshold algorithm must specify the direction of the inequality and plan a stopping condition.

Status and justification. Illustrative observations, without a formal theory of limits. The correctness of an integer threshold is checked by testing the index found and the previous index, when monotonicity is established.

Worked example. For $u_n = 80(\dfrac{1}{2})^n$, the first index such that $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The program begins with $n = 0, u = 80$ and repeats the division by 2 while $u \geq 6$.

Watch out

An unbounded loop may never stop if the threshold cannot be reached; $<$ and $\leq$ are not interchangeable.

Immediate application. For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.

Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 1.1 · Getting to grips with indices

Consider the sequence (u_n) defined for every natural number n by $u_n = 2n^2 - 3n$. The first term is therefore the one with index 0.

1. Calculate separately u_0, u_1 and u_4, by showing the substitution of n.
2. Express u_{n+1} as a function of n, then expand and simplify this expression.
3. Calculate $u_{n+1} - u_n$. By studying its sign for integer indices, describe the variation of the sequence.

Exercise 1.2 · Choosing the right model

A nursery currently has 100 young plants. It is considering two models to forecast its numbers: in model A, the number increases by 15 plants per month; in model B, it increases by 10 per cent per month. Let a_n and b_n the projected numbers after n months, with $a_0 = b_0 = 100$. The two models are studied separately.

1. **Model A.** Write the relationship between a_{n+1} and a_n. Identify the type of sequence, then express a_n as a function of n. Calculate a_2.
2. **Model B, after the study of geometric sequences.** Answer the same questions for (b_n) and calculate b_2.
3. For each model, explain what is assumed constant. Give one reason why a very long-term forecast could become unrealistic.

Exercise 1.3 · Recovering a sequence

The sequence (u_n) is geometric, defined from index 0. Its common ratio q is strictly positive. We know only $u_1 = 6$ and $u_3 = 54$.

1. Determine q by linking the two known terms. Show all the steps that make it possible to choose its value.
2. Deduce u_0, then u_5.
3. What would change if the hypothesis $q > 0$ were removed? Give all the possible common ratios and explain whether the value of u_5 would change.

Exercise 1.4 · Rows of seats

A concert hall has 12 rows. The first row has 18 seats; each following row has two more seats than the previous one. Let u_n be the number of seats in row n, for $1 \leq n \leq 12$.

1. Define the sequence (u_n) and calculate the number of seats in the last row.

2. Calculate the total capacity of the hall. Explain the number of terms used in the sum.

3. Calculate the mean number of seats per row and use this result to check the total found.

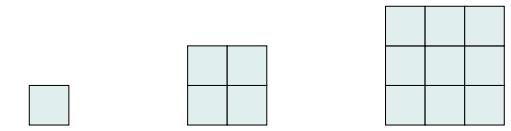
Exercise 1.5 · Adding up squares

Patterns are built with identical square tiles. Pattern number 1 contains one tile. To go from pattern number n to pattern number $n+1$, we add 2 $n+1$ tiles. Let u_n be the number of tiles in pattern number n, for $n \geq 1$.

1. Calculate u_2 and u_3. Draw an arrangement of these tiles that suggests a formula for u_n.

2. Conjecture an expression for u_n as a function of n.

3. Check that this expression gives the first term and satisfies the construction rule. Explain why these two checks identify the sequence.



Pattern 1 Pattern 2 Pattern 3

A possible layout for the first three patterns; the drawing helps to formulate a conjecture, which remains to be justified.

Exercise 1.6 · Two repayment plans

A person must repay 600 monetary units, interest-free. Two plans are being considered. With plan A, she pays 50 units at the end of each month until the loan is repaid. With plan B, she pays each month half of what she still owes. The calculations for model B are done without rounding. Let a_n and b_n be the debts remaining after the payment for month n, with $a_0 = b_0 = 600$.

1. Express a_n until the debt is paid off. In which month is the repayment completed?

2. Express b_n. Determine the first month in which the remaining debt is strictly less than 10 units ; check the previous month.

3. In model B without rounding, does the debt become exactly zero after a finite number of months? Justify and comment on the difference from a real situation.

Exercise 1.7 · Reading a list

A program must store the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It starts with the instructions `u=2` and `L=[u]`. It then executes, five times and in this order, `u=3*u-1` then `L.append(u)`.

1. Run the first two iterations of the loop by hand. Then give the contents of the list.
2. How many elements does the list contain after the five iterations? What is the index of the last term recorded?
3. Write the complete program in Python, then check the value of the last element. Explain why the number of terms is one more than the number of iterations.

Exercise 1.8 · Delayed review

This question is done one week after the chapter, without looking at the formulas. A sequence is defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate u_7 stating how many steps separate indices 2 and 7.
2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms of this sum and the number of steps calculated in the first question.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 64, questions a and b. Calculate the first four terms. Write the substitutions before simplifying.

CQFD, exercise 11 p. 64, all questions. Calculate the terms successively and keep a record of the four steps.

MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $u_n = 3n + 2$: u_4 value

   A. 14
   B. 18
   C. 12
   D. 6

2. $u_0 = 2$, $u_{n+1} = u_n + 5$: u_3 value

   A. 10
   B. 12
   C. 17
   D. 15

3. An increase of 8% gives the common ratio

   A. $0{,}08$
   B. $1{,}8$
   C. $0{,}92$
   D. $1{,}08$

4. From u_3 to u_9 included, we count

   A. 6
   B. 7
   C. 9
   D. 12

5. $1 + 2 + \cdots + 20$ is

   A. 200
   B. 420
   C. 210
   D. 190

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $u_n = -3 \times 2^n$ is

   A. strictly increasing
   B. constant
   C. neither increasing nor decreasing
   D. strictly decreasing

2. 1 + 2 + 4 + 8 equals

   A. 15
   B. 16
   C. 14

D. 8

3. For $u_n = \dfrac{16}{2^n}$, first index where $u_n < 2$:

A. 2

B. 3

C. 4

D. 5

4. The sequence $(-1)^n$

A. tends to 0

B. has no limit

C. is increasing

D. is constant

5. A list contains u_0 to u_{10} ; its length is

A. 10

B. 9

C. 12

D. 11

True or false? Justify.

1. Every geometric sequence with common ratio $q > 1$ is increasing.

2. If two sequences have equal first three terms, they are equal.

3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

Personal review

I distinguish explicit and recursive; I recognise a constant increment or rate; I count the terms; I check the index before a threshold.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

Solutions and teaching notes, chapter 1

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 11 : multiplication comes before addition. *Action:* Rewrite the steps of two calculations with brackets.
2. $1,\ 8,\ \frac{1}{4}$. *Action:* Build the chain of powers by dividing by 2.
3. $80 \times 1{,}05 = 84$. *Action:* Link 5 % to $\frac{5}{100}$ then to the coefficient 1.05.
4. -2 and 10. *Action:* Substitute x in brackets before calculating.
5. $10 - 3 + 1 = 8$. *Action:* Write out the list, then reconstruct the calculation.
6. $0, 1, 2, 3$. *Action:* Draw a two-column table, index and value; test range(1,4).

Investigation

Scheme A: last 45 kg, total $\frac{6(20+45)}{2}$ = 195 kg. B: $\frac{20(1-1{,}2^6)}{1-1{,}2} = 198{,}5984$ kg. B exceeds A in total, even if some initial terms are smaller. Calculator use is allowed here; argue using the cumulative total, not just the last term.

Direct applications

1. $2, 3, 6, 11$. The formula $w_n = n^2$ +2 works: it gives the first term and ($n+1)^2+2-(n^2+2) = 2n+1$.
2. $37 - 17 = 5r$, therefore r = 4 then $u_0 = 17 - 3 \times 4 = 5$.
3. $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. v_n is alternately 2 and -2: neither increasing nor decreasing.
4. 4 + 8 + 16 + 32 = 60, or $\frac{4(1-2^4)}{1-2} = 60$.
5. u_3 = 6 : index 3 for ≤ 6, index 4 for < 6.

Exercises: methods and answers

Exercise 1.1 - Getting to grips with indices. $u_0 = 0, u_1 = -1, u_4$ = 20. Expand with parentheses: $u_{n+1} = 2(n+1)^2 - 3(n+1) = 2n^2 + n - 1$; difference 4 $n - 1$. The sign changes between indices 0 and 1: the sequence first decreases, then increases from 1.

Exercise 1.2 - Choose the right model. $a_n = 100 + 15n$ and $b_n = 100(1{,}1)^n$, therefore $a_2 = 130$, b_2 = 121. A assumes a constant net addition; B a constant net rate. In reality, rounding and the carrying capacity limit these models.

Exercise 1.3 - Recovering a sequence. $u_3 = q^2 u_1$ so $q^2 = 9$. Positivity requires $q = 3$, hence $u_0 = 2$ and $u_5 = 486$. Without this detail, $q = -3$ would also be possible, with $u_0 = -2$ and the same $u_5 = 486$.

Exercise 1.4 - Rows of seats. The last row has $18 + 11 \times 2 = 40$ seats. Sum $\frac{12(18+40)}{2} = 348$. Check: a mean of 29 seats per row, between 18 and 40.

Exercise 1.5 - Adding up squares. We get 4 then 9 squares. A square with integer side n completed with a row and a column requires $2n + 1$ squares. The formula n^2 has the right first term and $(n+1)^2 - n^2 = 2n + 1$. The generation rule is the same.

Exercise 1.6 - Two repayment plans. For A: $a_n = 600 - 50n$ for $0 \leq n \leq 12$, then 0. For B: $b_n = \frac{600}{2^n}$. $b_5 = 18{,}75$ and $b_6 = 9{,}375$; the strict threshold is reached at index 6. B never gives exactly 0 in this ideal model: do not claim a finite repayment.

Exercise 1.7 - Reading a list. The first three elements are 2, 5, 14. An initial value and five appends give six elements; indices 0 to 5. The last one is $u_5 = 365$.

Exercise 1.8 - Delayed review. $u_7 = 12 + 5(-3) = -3$. Six terms: $\frac{6(12-3)}{2} = 27$. The 5 counts the steps, the 6 the terms.

CQFD, exercise 9 p. 64, questions a and b. a : $-1, 0, 3, 8$. b : $2, \frac{7}{2}, 4, \frac{17}{4}$. The first index is 0 ; the denominator of b is $n + 1$ and does not vanish on $\mathbb{N}$.

CQFD, exercise 11 p. 64, all questions. From $u_0 = 2$, the rule $u_{n+1} = 2u_n - 1$ gives $u_1 = 3$, $u_2 = 5$, $u_3 = 9$, $u_4 = 17$. The rule uses the previous term, not its index.

MCQ: answers and reasons

MCQ 1.

1. A. $3 \times 4 + 2 = 14$.
2. C. Three additions: $2 + 3 \times 5 = 17$.
3. D. The coefficient is $1 + \frac{8}{100}$.
4. B. $9 - 3 + 1 = 7$.
5. C. $\frac{20 \times 21}{2} = 210$.

MCQ 2.

1. D. The difference is $-3 \times 2^n < 0$.
2. A. Four terms; $\frac{1 - 2^4}{1 - 2} = 15$.
3. C. $u_3 = 2$, $u_4 = 1$.

4. B. The terms alternate between two distinct values.

5. D. Eleven indices from 0 to 10.

True or false

1. False: $u_n = -2^n$ is strictly decreasing.

2. False: $u_n = 0$ and $v_n = n(n-1)(n-2)$ coincide at indices 0, 1, 2, but not at index 3.

3. True: each term is 1 and there are n+1 terms.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.